JCIM JOURNAL OF CHEMICAL INFORMATION AND MODELING

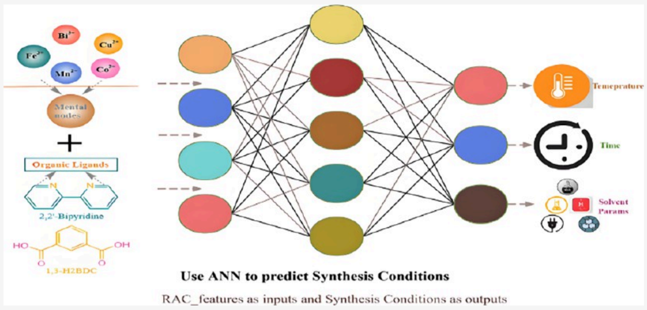
pubs.acs.org/jcim Article

An Intelligent Prediction Model for the Synthesis Conditions of Metal−Organic Frameworks Utilizing Artificial Neural Networks Enhanced by Genetic Algorithm Optimization

Guangying Jin,* Wei Ran, Manyue Zhang, and Yun Li

Cite This: J. Chem. Inf. Model. 2025, 65, 1085−1100

ABSTRACT: In the field of emerging materials, metal−organic frameworks (MOFs) have gained prominence due to their unique porous structures, showing versatility in gas adsorption, storage, separation, and liquid processes. However, their decomposition, collapse tendencies, and complex synthesis make large-scale production costly and challenging with no accurate method for predicting synthesis conditions. This work proposes an intelligent prediction model based on the structural characteristics of MOFs to forecast synthesis conditions. A genetic algorithm-optimized back-propagation (BP) neural network was developed, starting with feature selection via the minimum redundancy maximum relevance algorithm to rank feature importance. The optimal number of inputs and outputs was determined on the basis of performance, followed by genetic algorithm optimization of the BP neural network. The best initial population size and number of hidden nodes were identified. The study compared 10 models, including a genetic algorithm-optimized BP neural network and a simple BP neural network. The results revealed that the *R* coefficient of the optimized model reached 96.2%, surpassing that of conventional methods with all *R* values of approximately 85%. This approach allows for accurate prediction of MOF synthesis conditions, aiding material manufacturing in precise control over synthesis processes, improving material quality, and reducing raw material waste.



1. INTRODUCTION

Metal−organic frameworks (MOFs) are widely acclaimed as exceedingly promising porous substances. Originally conceived in the 1990s, Janssen et al.[1] introduced the notion of crafting durable and porous crystalline substances utilizing metal ions and organic ligands. These crystalline substances form structures with adjustable pore sizes and surface areas, exhibiting high efficiency in adsorbing, retaining, and separating gases, liquids, and various molecules. Over 20,000 different MOF materials have been synthesized by selectively utilizing different metal centers and organic ligands. Despite their wide range of applications, challenges persist in their utilization. For example, certain MOFs are susceptible to damage and decomposition under unstable conditions, such as fluctuating temperatures, humidities, and chemical environments.[2] The complex synthesis procedures also lead to the consumption of synthetic materials themselves, resulting in resource and financial waste. Furthermore, the performance of the MOFs is intricately influenced by their pore structure and function. Unfortunately, accurate prediction and optimization of these attributes remain elusive, particularly for most MOFs. Figure 1 illustrates several typical MOF preparation processes.

In Figure 1, RAC_features are utilized as inputs for the artificial neural network, where RAC_features encompass parameters such as geometric descriptors and chemical descriptors. The output comprises the conditions required for synthesizing MOFs, including temperature (a single temperature, rather than multiple temperatures), time, and solvent parameters (such as the octanol/water partition coefficient, boiling point, melting point, etc.). Many laboratory studies on the synthesis of MOFs mention that the same MOF material has different synthesis conditions, but the purpose of this work is to predict one of the conditions that can be used to successfully synthesize MOF materials. By training the artificial neural network is trained with these parameters, the relationship between the input and output can be discerned, leading to the identification of the optimal model. Additionally, Figure 1 also delineates the characteristic parameters designated as inputs and

Received: June 10, 2024
Revised: December 24, 2024
Accepted: January 8, 2025
Published: January 20, 2025

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

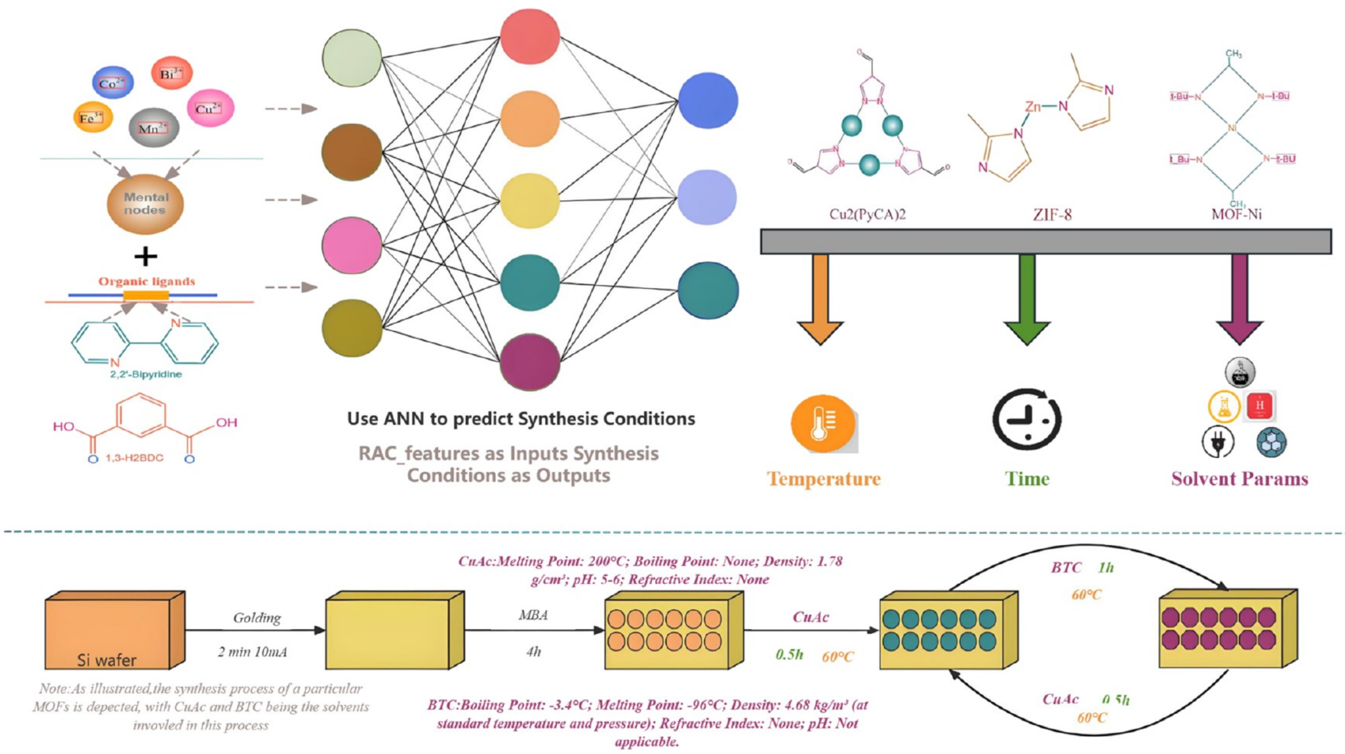


Figure 1. Schematic of the study content: RAC_features (including parameters such as geometric descriptors and chemical descriptors) as inputs to the neural network, the synthesis conditions of MOFs (including temperature, time, octanol/water partition coefficient, etc.), and the synthesis conditions involved in some common synthesis processes.

outputs. While conventional synthesis methods such as hydrothermal and microwave synthesis are used, they often lead to inaccuracies and waste raw materials and funds. Despite attempts to use machine learning models for synthesis control, they still lack precise predictions and contribute to waste. Thus, more accurate prediction methods are urgently needed to better anticipate synthesis conditions, enhance control over material synthesis parameters, and reduce resource waste.

Therefore, in response to these challenges, this paper introduces a novel approach using neural networks and revised autocorrelation (RAC) features, which refers to the feature vector that combines the geometric descriptors and chemical descriptors of MOFs, including pore size, pore volume, surface area, metal nodes, organic ligands, and other characteristics, structural features to predict optimal synthesis conditions for MOFs, such as the temperature, time, and solvent parameters required for synthesizing MOFs, such as the boiling point and distribution coefficient, to enable controlled synthesis and customized design. ZEO++ is an open-source software package designed for molecular simulations and computational chemistry. It is utilized primarily for molecular dynamics (MD) and Monte Carlo (MC) simulations, offering efficient computations for large-scale systems. ZEO++ enables the analysis of crystal structures and molecular arrangements and is primarily used for calculating the geometric characteristics of materials. Its structure is similar to that of several studies, such as CIF, and is used to describe the coordination environment of the coordination centers in MOFs and the coordination relationship between the metal centers and organic ligands. Thus, BP neural networks are utilized to forecast the synthesis conditions of MOFs, encompassing parameters such as temperature, time, and solvent characteristics, including the melting point, boiling point, octanol/water partition coefficient, polarity, and pH based on RAC_features (chemical descriptors of MOFs) and pore-geometric features (pore-geometry descriptors of MOF calculated by Zeo++). This method has the potential to reduce waste and facilitate large-scale production for various production enterprises.

This paper is structured as follows: Section 2 covers traditional MOF synthesis methods, challenges, and relevant intelligent methodologies. Section 3 discusses the MOF structure and application methods, and Section 4 presents the research findings. Finally, Section 5 explores the conclusions, discussion, and future development directions.

2. LITERATURE VIEW

MOFs, which are renowned for their nanoporous structure, are prized for their adsorption and catalytic capabilities, which are driven by their high porosity, extensive surface area, structural diversity, and adaptable metal coordination sites.[3] Despite these advancements, achieving flawless MOFs remains challenging owing to the sensitivity of material defects to minor changes in the reaction conditions. This highlights the need for thorough study and management to control material traits despite ongoing efforts to produce defect-free MOFs.

However, in traditional MOF synthesis, these materials are prized for their adsorption and catalytic abilities due to their nanoporous structure, high surface area, and structural diversity. They are widely used in various fields, notably in supercapacitor materials.[4] However, perfect MOFs are difficult to attain because of their sensitivity to the reaction conditions, resulting in defects. Understanding defects is crucial for controlling material properties.[5−7] The detection and synthesis of defective MOFs are still being advanced. Traditional synthetic methods often require repetitive experimentation due to the material properties. This approach results in significant waste of raw

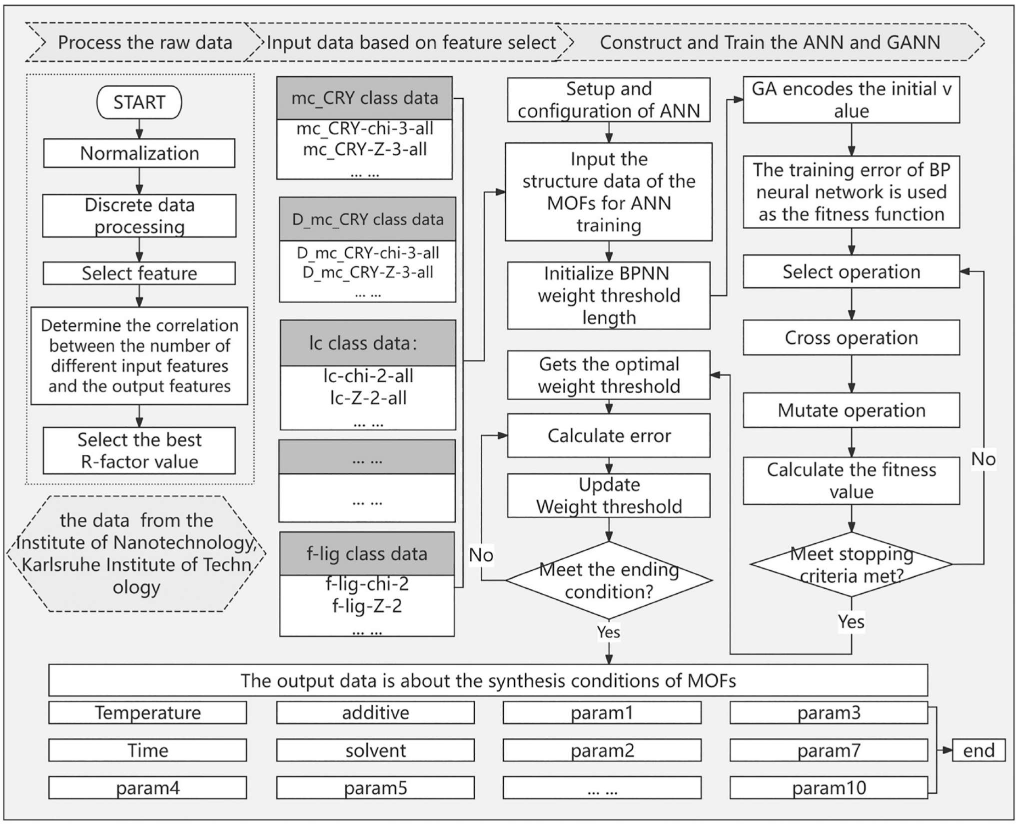


Figure 2. Whole process of the intelligent metal−organic framework prediction model.

materials. Zhang et al.[8] employed a straightforward thermal solvent method for MOF synthesis, and Yang et al.[9] synthesized nickel-based MOFs for multiple cycles of experiments. This undeniably leads to substantial material waste. Additionally, in some scholarly research, variable control and adjustment of proportions and reaction conditions for MOF synthesis are employed,[10] epitomizing a trial-and-error approach that demands considerable input of experimental materials. Moreover, extensive research has been undertaken by various scholars focusing on the chemical stability of MOFs. For example, McHugh et al.[11] and Ding et al.[2] investigated the stability of MOFs under hydrolytic conditions and proposed corresponding solutions. Conventional synthesis methods of MOFs often result in material waste and financial inefficiencies due to inaccuracies. Therefore, material manufacturing enterprises (chemical enterprises, transportation equipment enterprises, and so on) urgently require a method for precisely forecasting material synthesis conditions, aiming to mitigate raw material waste and capital inefficiencies.

Moreover, in this information age, neural networks have been proven to be effective means of prediction and have been widely applied across a multitude of fields. Prominent models include linear regression, logistic regression, decision tree, random forest, support vector machine (SVM), naïve Bayes, and neural networks. Research frequently juxtaposes their efficacy; for example, Ding et al.[12] reported that artificial neural networks surpassed decision trees and XGBoost in prognosticating maximum power density. In a separate inquiry,[13] artificial neural networks demonstrated superior performance over multiple regression in forecasting saccharification rates during biomass acid pretreatment. Similarly, investigations comparing SVM and ANN in stock prognostication[14] revealed heightened accuracy with ANN. Overall, neural networks exhibit heightened precision in prognostication across diverse domains.[15]

Therefore, as an artificial intelligence method, machine learning can make precise predictions and can be applied in various fields. In accordance with Timoshenko et al.,[19] neural networks have been employed to scrutinize the structure and chemical transformations of spinel-shaped catalysts. Their NN-EXAFS investigation revealed the intricacies and heterogeneities occurring during the activation treatment and the OER process of $Co_xFe_{3-x}O_4$ catalysts. Furthermore, neural networks have been applied in X-ray-related endeavors. In the study of Kotobi et al.,[20] various graph neural network architectures have been constructed to compare the predictive performance of XAS spectra. In the research by Illarionov et al.,[21] a proposal was proposed to amalgamate analytical polarizable force fields with short-range neural network corrections for analyzing intermolecular interaction energies. Additionally, numerous articles currently employ machine learning in synthesizing MOFs, addressing the challenge of planning their synthetic routes efficiently, as highlighted by Lin et al.[22] Raccuglia et al.[23]trained a machine learning model on failed hydrothermal reaction data to predict new reactions, achieving an 89% success rate in guiding the synthesis of metal−organic materials. Machine learning can be used to optimize synthesis parameters, transforming the MOF field. Xie et al.[24] predicted metal−organic nanocapsule crystallization with over 90% accuracy via 486 data points. Zhang et al.[25] analyzed the particle size variation in Eu-MOFs, aiding in the evaluation of synthesis conditions. Vo et al.[16] trained neural networks with low computational costs on synthetic data, demonstrating the accuracy and versatility of artificial neural networks in prediction tasks. Research on the synthesis of MOFs suggests that the same MOF material can be

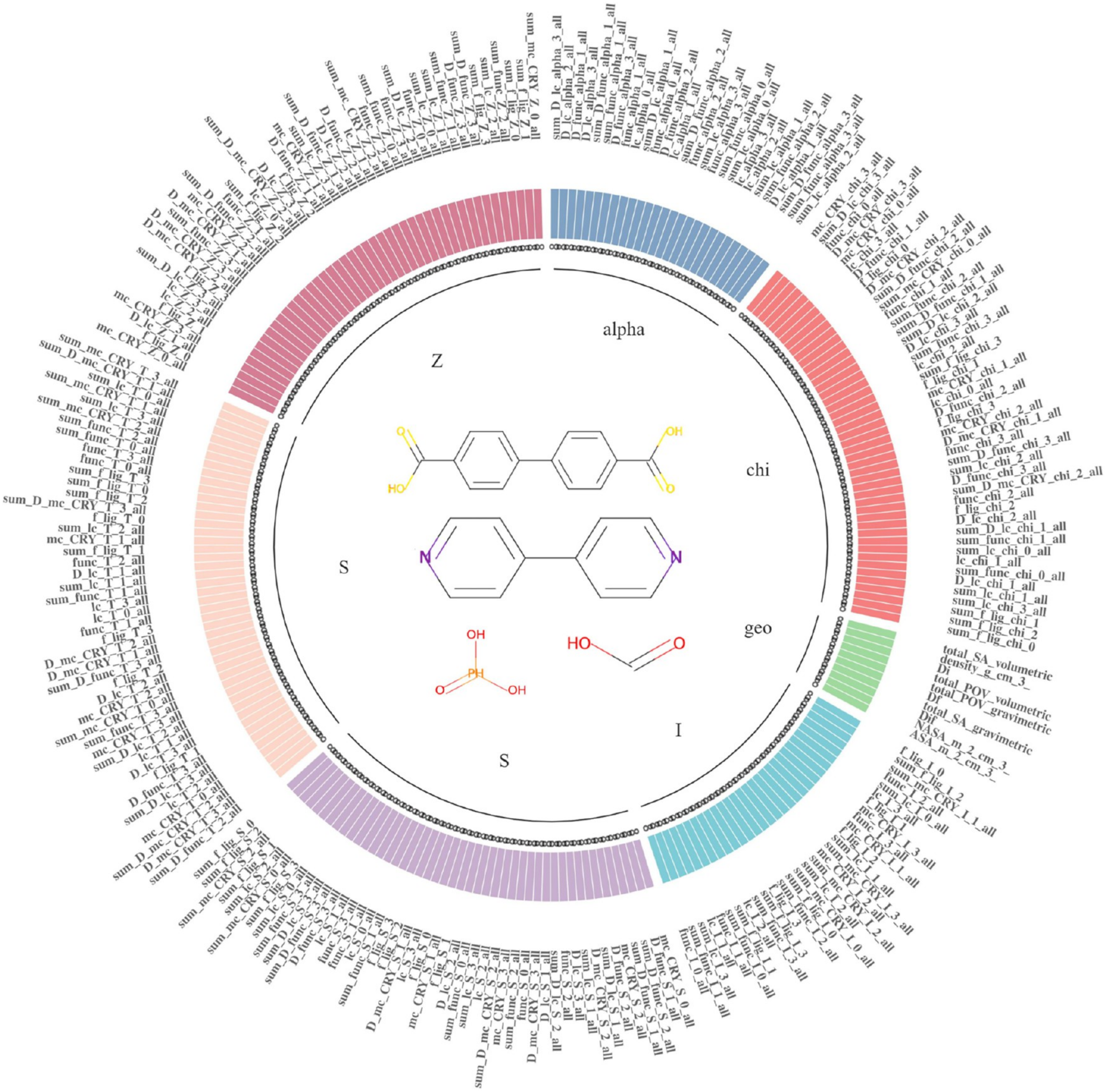


Figure 3. All unprocessed raw input feature parameters.

synthesized under different conditions, but the purpose of this study is to use machine learning to predict which combination of conditions can be used to predict the MOF material successfully.[17,18]

Therefore, artificial neural networks excel in prediction but face challenges such as parameter selection, overfitting, and slow convergence, and they can study the synthesizability of materials from a higher dimension. Meanwhile, in Luo et al.'s study,[26] the synthesis conditions of MOF were predicted utilizing crystal information files. Although the results were not ideal, they indicated that the outcomes were significantly superior to expert intuition, serving as a significant driving force for the research direction of this paper. Genetic algorithms offer global search capabilities and adaptability, optimizing parameters and structures effectively.[27] Moosavi et al.[28] and others have used genetic algorithms to optimize MOF synthesis conditions, increasing the surface area production and reaction time. This finding underscores the potential of genetic algorithms for intelligently synthesizing MOFs and efficiently navigating complex parameter spaces for optimal combinations.

Moreover, past research has emphasized the effectiveness of genetic algorithms in MOF synthesis and the advantages of artificial neural networks. However, the prediction of the MOF synthesis conditions is still not accurate. Through the investigation of traditional synthesis methods in the second paragraph of this section, it has been revealed that such approaches result in material waste and financial pressures. Furthermore, the studies conducted by Zhang et al.[29] and Yang et al.[30] have also assessed the material and financial waste inherent in traditional synthesis methods. Many experts are

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

actively seeking more efficient synthesis techniques to address these issues. Therefore, to address these challenges, a new method using neural network optimization and a genetic algorithm has been proposed, aimed at predicting the synthesis conditions of MOFs on the basis of RAC structural features and reducing material waste and cost for manufacturing enterprises.

3. MATERIALS AND METHODS

As depicted in Figure 2, the primary framework establishment process for this manuscript encompasses the preprocessing of raw data, model construction, and subsequent steps.

According to the architectural diagram (Figure 2), at the onset of model establishment, preprocessing of the raw data is imperative. This study involves data normalization, discrete data handling, and feature selection prioritizing highly correlated input features. An artificial neural network is then used to optimize the number of input and output features, which are refined through genetic-algorithm-based optimization including population initialization and the determination of hidden layer node counts. The optimal model is identified through evaluation metrics.

3.1. Data Collection and Structure of the MOFs. Owing to the complexity and sensitivity of MOF structures to synthesis conditions, limited data are available. This study utilized 1341 data sets from the material laboratory of the Karlsruhe Institute of Technology,[31] incorporating structural features and synthesis conditions from the published literature. In this data set, the temperature distribution ranges from 5 to 220 °C, the time distribution spans from 5 to 672 min, and the distributions of the solvent parameters and other synthesis conditions are random. The model was trained on existing research data, with 1000 pieces used for training and 341 for testing, and in this article, a hiplot was used to generate a display image of the raw data. Figure 3 illustrates the untreated input feature parameters, totaling approximately 251. As illustrated in Figure 3, all feature variables are classified into seven categories, with geometric descriptors being one category and chemical descriptors being six categories. The specific contents of the chemical descriptors are listed in Table 1.

To better explain these variables, this review categorizes all feature names on the basis of different operations (product and difference). RACs include four possible initiations and two possible scopes: the metal center (mc) start, the linker ligand atom center (lc) start, the functional group center (func) start, each atom (full) start, all atoms in the original unit (all) scope, or all atoms in the linker (linker) scope. All starts, ranges, and operations use weld depths of 0, 1, 2, and 3 to generate autocorrelations and then classify them on the basis of different heuristic properties weights: nuclear charge (Z), electronegativity (chi), topology (T), identity (I), valence radius (S), and unsaturation (alpha), and feature names are classified on the basis of different heuristic property weights. RAC features using symbols are indicated by their beginning (e.g., mc), and RAC features using differences contain a "D" prefix that begins with a subscript (e.g., D_mc). For example, mc-S-0-all represents the feature names of product operations with different covalent radii (heuristic properties quantities) starting from the metal center (start), and D_mc-I-all represents the feature names of difference operations with different identities (heuristic properties quantities) starting from the metal center (start).[28,32]

Table 1. Designation of Each Category of Features and the Significance of Each Classification

operation	start	heuristic properties	the meaning of each heuristic property	examples of RAC features
product	mc	T	connectivity	mc-S-0-all
		Z	nuclear charge	mc-S-1-all
		alpha	polarizability	lc-alpha-1-all
	lc	chi	electronegativity	lc-alpha-0-all
				func-I-2-all
	func	I	identity	func-I-3-all
	full	S	valence radius	func-S-0-all
difference	mc	T	connectivity	D_mc-T-2-all
		Z	nuclear charge	D_mc-T-1-all
		alpha	polarizability	D_func-S-0-all
	lc	chi	electronegativity	D_func-S-2-all
				D_lc-I-2-all
	func	I	identity	D_lc-chi-0-all
	full	S	valence radius	D_lc-chi-2-all

The synthesis of MOFs typically involves the use of metal ions or clusters and organic ligands as raw materials. Commonly used metal ions and aluminum, metal clusters can also be used. Organic ligands are compounds with organic groups that connect metal ions in the MOF synthesis. These ligands often have multiple coordination sites for bonding with metal ions. Solvents, such as water, DMF, and DMSO, are used to create a suitable reaction environment and dissolve the raw materials. Proper reaction temperature and time are also crucial for promoting coordination reactions and MOF growth.

In summary, the synthesis of MOFs generally involves the use of metal ions or clusters and organic ligands as the main raw materials combined with solvents and appropriate reaction conditions. By selection of the right raw materials and conditions, MOFs with different structures and properties can be synthesized.

3.2. Feature Selection Algorithm. Feature selection is paramount in feature engineering; it identifies the optimal subset of features by eliminating irrelevant or redundant features, thereby increasing the model accuracy and reducing the computational burden. Selecting pertinent features simplifies the model, aids in understanding the data generation process, and mitigates the risk of overfitting. Feature selection is an essential tool in machine learning and data mining for identifying the most predictive features. This study employs the minimum redundancy maximum relevance (MRMR) algorithm to minimize feature redundancy and maximize the relevance to the target variable, thereby selecting the optimal subset of features:

1. Maximize relevance: The MRMR algorithm selects features that are highly correlated with the target variable by calculating their correlation, typically via Pearson correlation coefficients or mutual information.
2. Minimize redundancy: The MRMR algorithm prevents the selection of highly correlated features by considering feature redundancy, quantified through correlation calculations such as Pearson correlation coefficients and mutual information.
3. Select the best subset of features: The MRMR algorithm selects the optimal subset of features by evaluating the difference between each feature's correlation and its redundancy with other selected features, prioritizing those with the highest variance. The following is an overview of the MRMR algorithm:

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

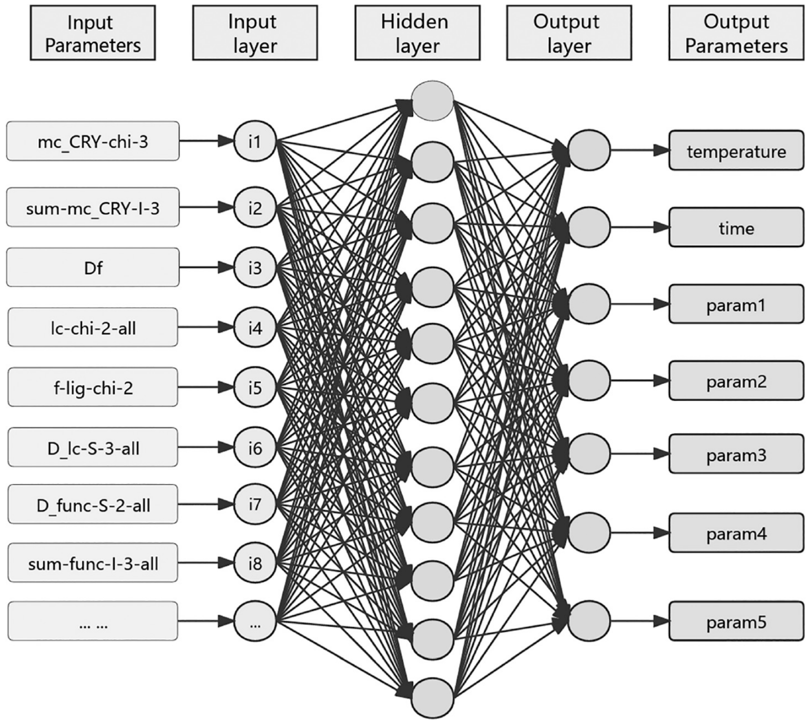


Figure 4. Schematic diagram of the neural network.

However, since max-dependency is difficult to implement, a generational alternative is max-relevance, which satisfies the characteristics of the following formula:

$$\max D(S,\ c),\ D\frac{1}{|S|}\sum_{x_i \in S} I(x_i;\ c) \tag{1}$$

The average mutual information between each feature x_i and classification c approximates $D(S, c)$. The features selected via max-relevance may exhibit a high dependence, often leading to redundancy. The removal of one redundant feature minimally alters the classification structure, necessitating the use of the min-redundancy method for elimination, as per the following formula:

$$\min R(S),\ r\frac{1}{|S|^2}\sum_{x_i, x_i \in S} I(x_i,\ x_j) \tag{2}$$

The combination of the maximum correlation D and the minimum redundancy R is called the non-minimal-redundancy-maximum-relevancy (MRMR). The method $\varphi(D, R)$ can be defined as the combination of D and R, considering the simplest method of combination:

$$\max \phi(D,\ R),\ \phi = D - R \tag{3}$$

In practical applications, the approximate optimal solution can be obtained using the defined operator φ and the incremental search method. Assuming that the S_{m-1} feature has been obtained, the feature needs to be selected in the subset of the features of $X - S_{m-1}$, the feature is selected by maximizing φ, that is, maximizing, and the calculation formula I is as follows:

$$\max_{x_j \in X - S_{m-1}}\left[I(x_j;\, c)\frac{1}{m-1}\sum\nolimits_{x_i \in S_{m-1}} I(x_i;\, x_j)\right] \tag{4}$$

3.3. Artificial Neural Network. The earliest concept of artificial neural networks can be traced back to a paper[33] written in 1943 by the neuroscientist Warren McCulloch and mathematician Walter Pitts. This paper introduces a simplified model of biological neurons, demonstrating their ability to simulate logical operations and marking a foundational contribution to ANN research. It pioneers the integration of biological neuron principles into computational models, setting the stage for subsequent neural network studies.

An artificial neural network mimics the structure and operation of a biological neural network, which consists of multiple artificial neurons (or nodes) that are connected to each other by connections or weights. ANNs are basically divided into multiple layers, including the input layer, the hidden layer, and the output layer. Each neuron receives input from the upper layer of neurons, weights the summation of the inputs, and transmits the results to the next layer of neurons via an activation function. His training process is achieved through a back-propagation algorithm, which adjusts the connection weights through formulas so that the output of the network is as close to the true value as possible. After training is complete, artificial neural networks can be used for tasks such as prediction and classification, and recognition ANNs have a wide range of applications in various fields, including image recognition, natural language processing, financial forecasting, and medical diagnosis. With the development of deep learning, deep neural network structures (such as convolutional neural networks and recurrent neural networks) have gradually become an important part of the field of artificial intelligence. The approximate neural network model used in this study is presented in Figure 4, and the main input and output feature indicators are listed in Table 2.

The artificial neural network described here comprises multiple inputs and outputs. Table 2 shows that the input included the MOF crystal structure, whereas the output included synthesis conditions such as temperature, time, and solvent parameters (such as the distribution coefficient, point, melting point, etc.). Through training, the model predicts

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

Table 2. Input and Output Characteristics of the Raw MOF Data

input features	output features
mc_CRY-chi-3-all	temperature
mc_CRY-S-3-all	time
sum-D_mc_CRY-chi-3-all	Param1
Df	Param2
lc-chi-2-all	Param3
......	......
lc-Z-2-all	Param10

synthesis conditions on the basis of the MOF crystal structure. This study assesses various configurations of inputs, hidden layers, and outputs to determine the optimal combination by comparing R coefficients across models.

3.4. Artificial Neural Network Optimized via the GA. The genetic algorithm (GA) is an optimization algorithm based on natural selection and evolutionary processes.[34] Once the ANN model has been constructed, integrating it with the GA tends to produce optimal values for all input parameters that lead to all the best synthesis results. The genetic algorithm is actually a heuristic search method borrowed from Darwin's theory of evolution and is used to solve the optimization problem.[35] The algorithm is based on the concept of populations, each containing multiple individuals (or chromosomes) representing a possible solution to the problem. Genetic algorithms mimic processes such as natural selection, crossover, and mutation to progressively optimize individuals in a population to arrive at better solutions. The basic steps of the genetic algorithm include the following steps:

1. Initialize the population: A certain number of individuals are randomly generated as the initial solution of the population.
2. Selection: Individuals are evaluated according to the fitness function, and individuals with higher fitness values are selected as parents.
3. Crossing: Crossover operations is performed on the selected parents to produce offspring.
4. Mutation: Mutation operations are performed on the offspring to introduce new changes.
5. Renewing the population: According to a certain selection strategy, the offspring are added to the original population, and the individuals with lower fitness are discarded.
6. The selection, crossover, mutation, and updated steps are repeated until the stop condition is met.

Its basic flowchart is shown in Figure 5.

3.5. Model Validation. Model validation is a foundational method of machine learning that evaluates the performance of a machine learning model on new data either internally or externally. The initial practice of internal validation of the model is to randomly divide the model development cohort into a training set, which is used for model fitting. The validation set is used to evaluate the performance of the model fitted by the training set, and the data are divided into two parts for "internal validation".

The model validation queue is the model validation queue, which is an external validation queue independent of the model development queue and uses data that have not been used in model development to evaluate the model's performance on new data. Compared with internal validation, external validation is more concerned with the portability and generalizability of the

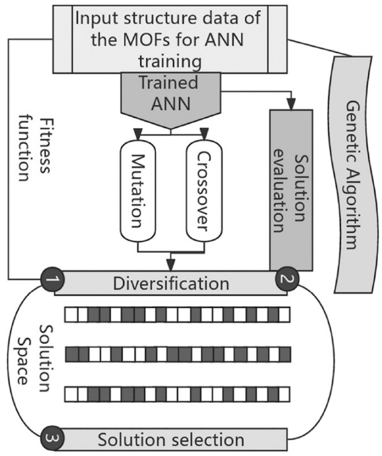


Figure 5. Processing flow of the genetic algorithm

model; that is, whether the performance of the model in different time periods, different regions, or different populations from the model development cohort is consistent with model development. To improve the quality of the research results and make the prediction model more credible, the developed model is externally validated after model development and internal validation are completed and the validation results are reported together with the model development process in the study. In addition, it is possible to externally validate the published model of others using existing data or as a separate model for external validation.

3.6. Evaluation Indicators. The neural network diagram Pearson correlation coefficient analysis was used to analyze the correlation between the two variables. The Pearson test is used to evaluate the correlation between the fitting results and the original data. The value of the correlation coefficient (R) can vary between (−1, 1). A value of 1 indicates a complete positive correlation, which means that the tested data set has an absolute linear relationship. R is used to quantitatively analyze the accuracy of the fitting results (Liu and Chen, 2020). To evaluate the predictive performance of the model on testing, validation, and prediction data, the Pearson correlation coefficient (R) and mean square error (MSE) are considered appropriate accuracy measures and are defined as follows:

$$R = \frac{\sum (G_m - \bar{G})(K_m - \bar{K})}{\sqrt{\sum (G_m - \bar{G})^2 (K_m - \bar{K})^2}} \tag{5}$$

$$\mathrm{MSE} = \frac{\sum (G_m - K_m)^2}{I} \tag{6}$$

where G_m represents the value of the observed data, K_m represents the predicted value, and I represents the length of the data. $\bar{G}$ represents the average value of the observed data, and $\bar{K}$ represents the average value of the predicted data. Typically, models with larger R values and smaller MSE values perform better, meaning that smaller MSE values lead to higher performance; the closer the R value is to 1, the higher the performance.

4. RESULTS

In this section, the results of the crystal structure prediction synthesis conditions based on the MOFs are analyzed, including observations of the data, processing, prediction of the synthesis conditions, optimization results, and problem solution reports.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

4.1. Feature Selection Determines the Number of Inputs. In this study, with a data set containing up to 340 initial input features, feature selection is crucial for improving predictive performance and preventing overfitting. Many of these features share similar meanings, which could adversely impact the model performance.

To address this issue, we used the MRMR algorithm for dimensionality reduction via feature selection. We then analyze various machine learning models' performances with different feature selections on the basis of feature count. Our results show that neural network models perform better with feature-selected data sets, likely because noise features are minimized, and underlying patterns in the data are highlighted.

In Figure 6, the average scores of each class are depicted. Notably, the average importance values were greater for the geo class (0.1408), alpha class (0.0755), and S class (0.07217), suggesting their significant impact on predicting final synthesis conditions compared with other feature types.

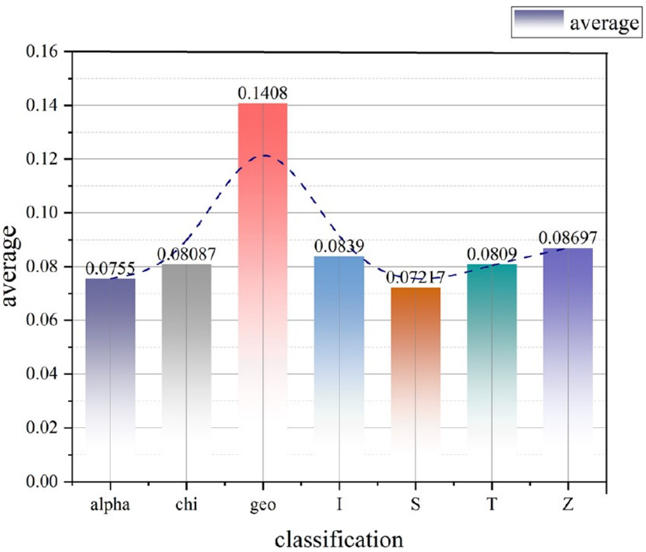


Figure 6. Average importance score of each class.

The features with an importance score of 0 are eliminated according to their importance ranking. The remaining features are grouped on the basis of their structural attributes, and their average importance scores are illustrated in Figure 6. Each category comprises input features with high importance scores, maintaining consistent rankings across categories.

4.2. Analysis of Artificial Neural Network Results. The artificial neural network (ANN) has a wide range of applications in the field of machine learning, and considerable attention has been given to the impact of selecting the number of input features on its performance.

In this study, a common feature selection algorithm (MRMR feature selection algorithm) was used to determine the number of different input features after which the artificial neural network was trained and tested. We conducted experiments on multiple data sets by comparing R values to ensure the optimal number of inputs.

According to Figure 7, the number of different input features significantly affects the performance of the ANN. When the number of input features is small, the generalization ability of the network decreases, and when the number of input features is large, the network is prone to overfitting. Figure 7 shows that when the number of input features is 123, the R values are in a relatively optimal state. Finally, the optimal number of inputs for the neural network is determined to be 123 as shown in Figure 8.

Simultaneously, a comparative analysis of the MRMR feature selection algorithm employed in this study was conducted against various stochastic feature selection methods (such as the Pearson feature selection and Spearman feature selection algorithms), with the results illustrated in Figure 9.

As illustrated in Figure 9, a comparative analysis was conducted between the selected random feature selection algorithm and the MRMR algorithm. Pearson feature selection involved, and Spearman feature selection underwent regression analysis with MRMR feature selection for the same number of features. The results revealed that their R values were better when the number of features was 123; however, the R value of the MRMR algorithm surpassed that of both the Pearson and Spearman feature selection methods. Moreover, in order to verify that the MRMR algorithm is the chosen feature algorithm with the highest accuracy, as shown in Figure 10, this paper

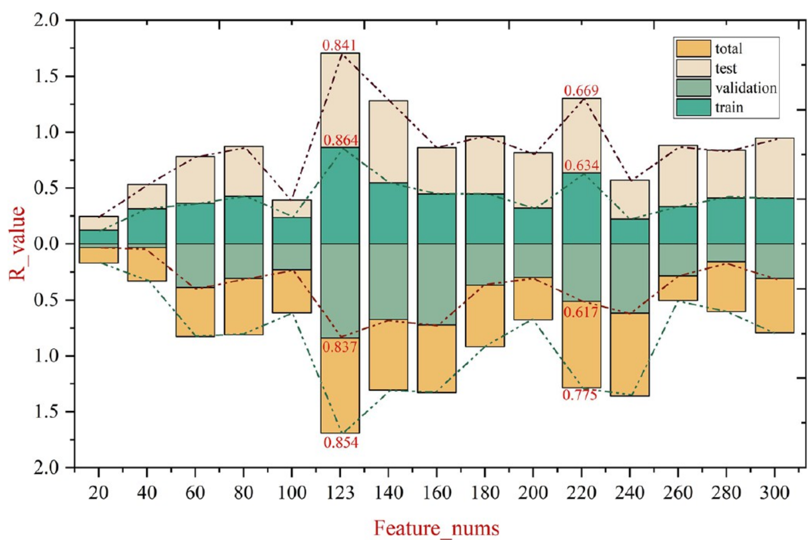


Figure 7. Comparison of R values for different input feature numbers.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

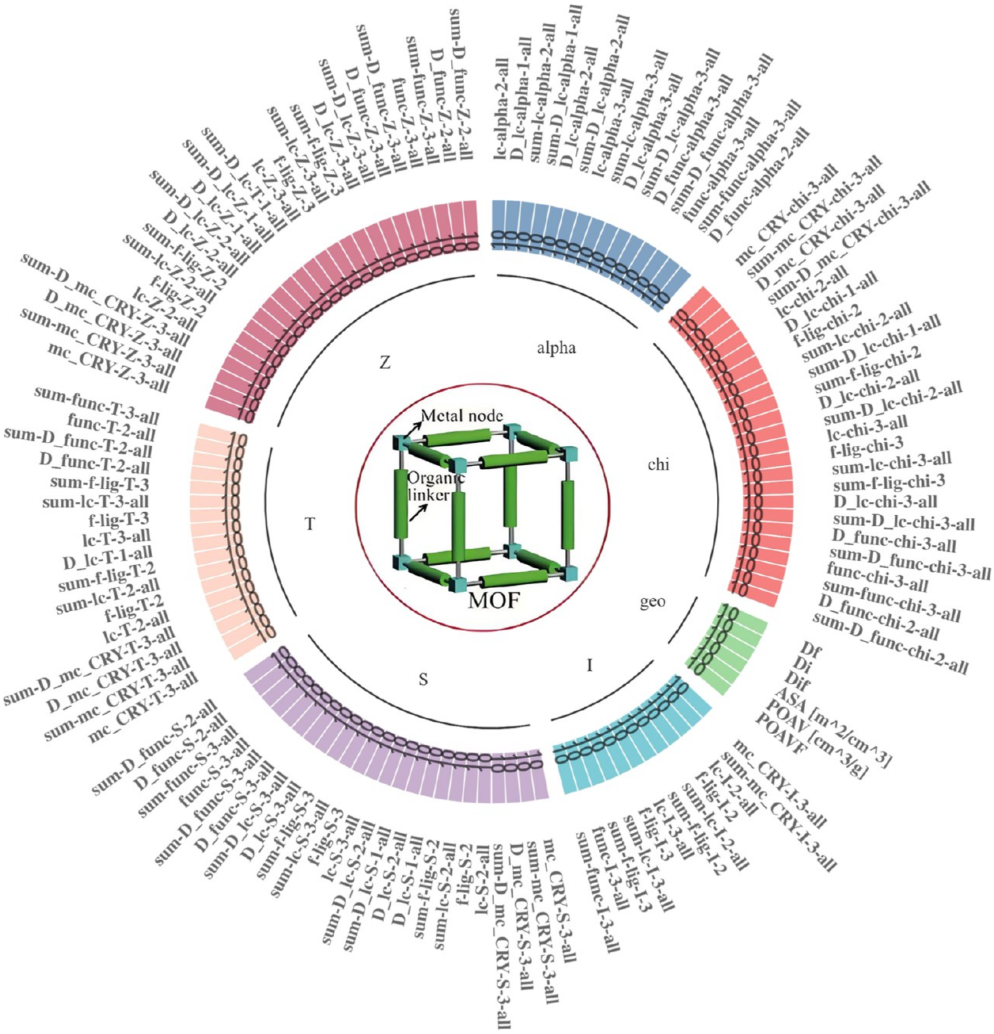


Figure 8. Data pertaining to the optimal input count after rigorous screening.

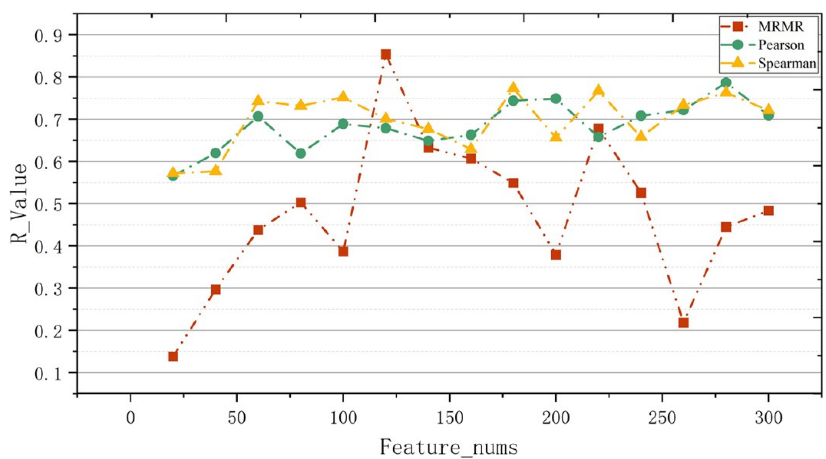


Figure 9. Comparative analysis chart of random feature selection methods with MRMR regarding fitting value *R*.

compares the model results of these three feature selection algorithms on 123 feature counts with respect to the number of training iterations. The results show that the MRMR algorithm is committed to finding the optimal model with the best performance.

After the optimal number of inputs was determined, the training model revealed that the regression effect was still poor, so the output features were judged. The model is trained with a single output feature as the output of the neural network, the training result is observed, and the output feature with an *R* value greater than 0.5 is used as the final output, as shown in Table 3.

In summary, 123 optimal input features are obtained, and 7 optimal output features are obtained. A comparison between the

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

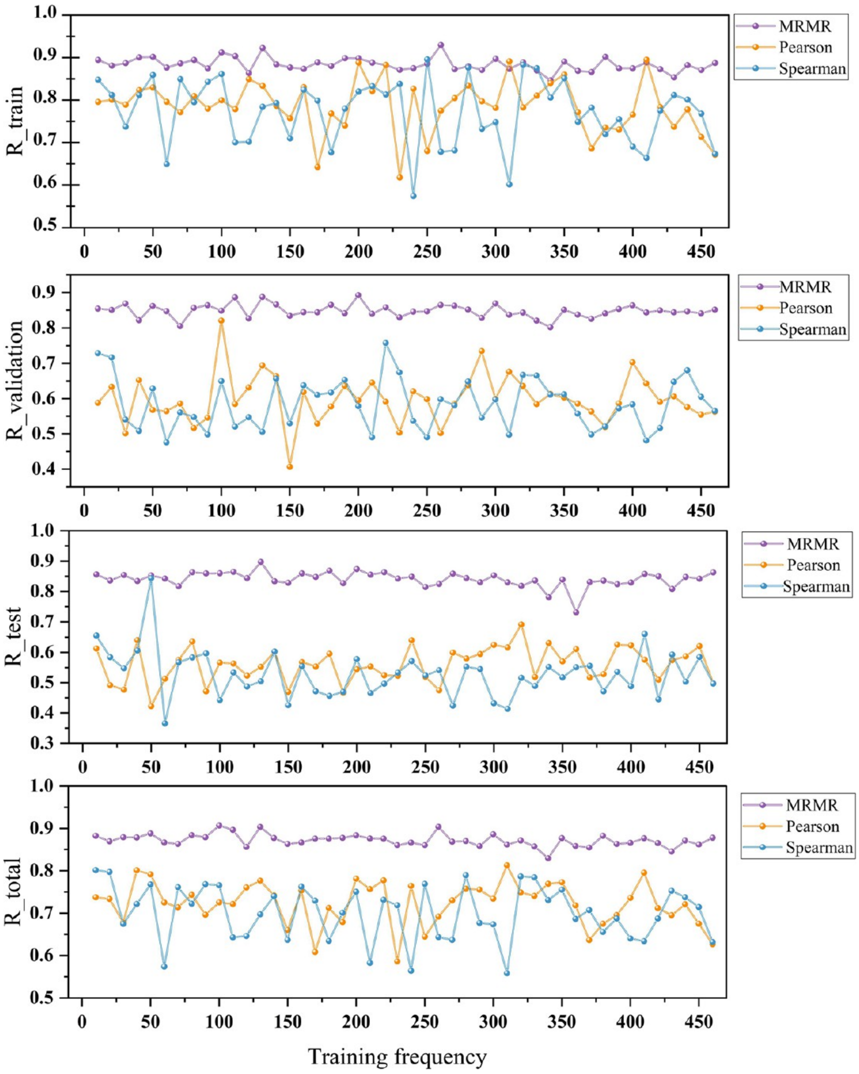


Figure 10. Comparison and analysis of the performance of different feature selection algorithms on 123 features with different training frequencies on the basis of the R results.

Table 3. Correlation Analysis of the Input Features and Single Output Feature[a]

	index			
feature	R (train)	R (validate)	R (test)	R (total)
temperature	0.92143	0.91445	0.89	0.87
time	0.85412	0.87412	0.77	0.8
Param1	0.76952	0.74077	0.53	0.59
Param2	0.90911	0.84381	0.8	0.75
Param3	0.91513	0.90868	0.81	0.81
Param4	0.95938	0.93204	0.9	0.87
Param5	0.55918	0.49939	0.34	0.3
Param6	0.52447	0.42827	0.21	0.19
Param7	0.64621	0.54166	0.38	0.33
Param8	0.59196	0.52254	0.3	0.27
Param9	0.63714	0.44656	0.37	0.28
Param10	0.79585	0.72577	0.61	0.57

[a]Note: The specific meanings of Param1 through Param10 are as follows: octanol/water partition coefficient, number of hydrogen bond donors, number of hydrogen bond acceptors, maximum absolute value of local charge, boiling point, melting point, pH value, density, refractive index, and polarity.

initial data and the processed data after training with the same model is shown in Figure 11.

Therefore, processing the input and output features of the original data can improve the fitting performance of the model. Notably, the Pearson correlation coefficient R of the original data (training set: 0.8066, validation set: 0.78556, testing set: 0.79375, all: 0.80295), after processing the input and output features, increased to R (training set: 0.87192, validation set: 0.84371, test set: 0.84645, all: 0.86385), which is conducive to improving the prediction accuracy of the synthesis conditions.

4.3. Results Analysis of the ANN Optimized by the GA.

4.3.1. Parameter Adjustment of the Neural Network. Genetic algorithm optimization is widely used in artificial intelligence. Here, we apply it to optimize artificial neural network parameters and analyze the results. Our experiments show that genetic algorithms improve network performance across multiple data sets by exploring key optimization factors. This approach mimics nature's evolution process, efficiently finding global optimal solutions, thus reducing the complexity and time involved in parameter optimization for neural networks. This study aims to investigate the application of genetic algorithm optimization in neural networks and its impact on performance.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

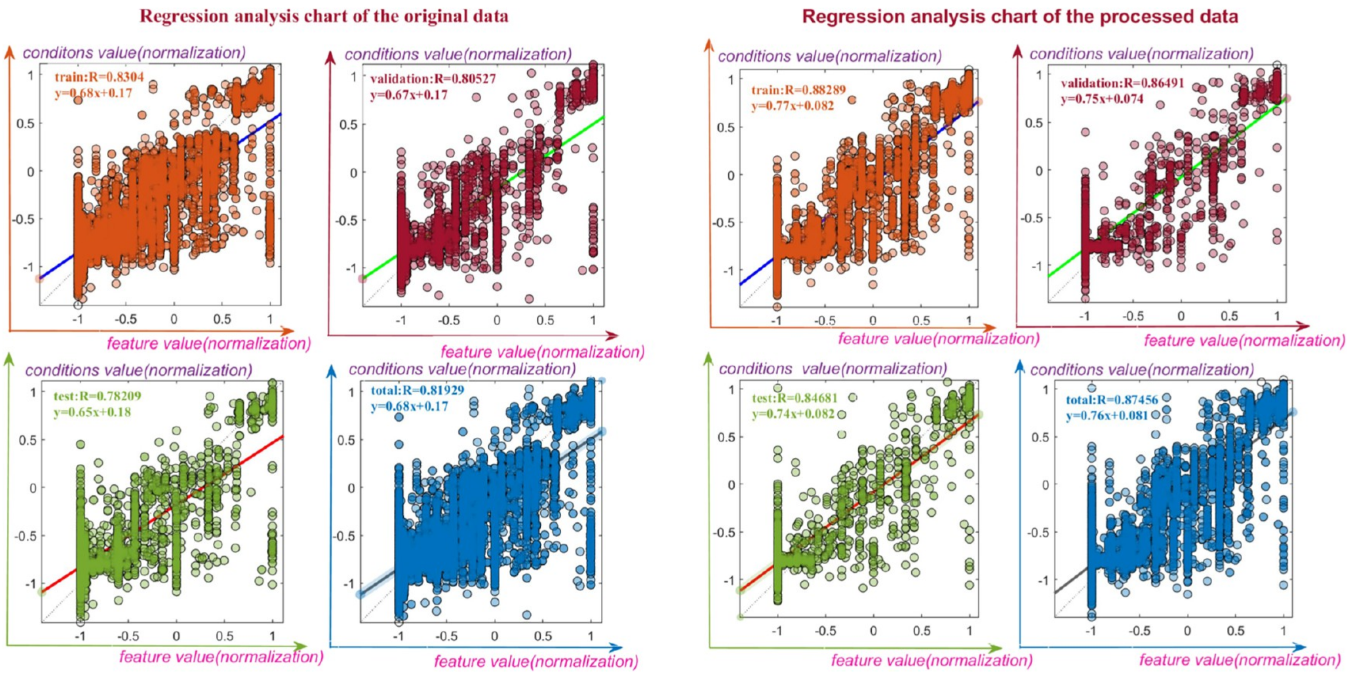


Figure 11. Regression fitting analysis of the raw and processed data.

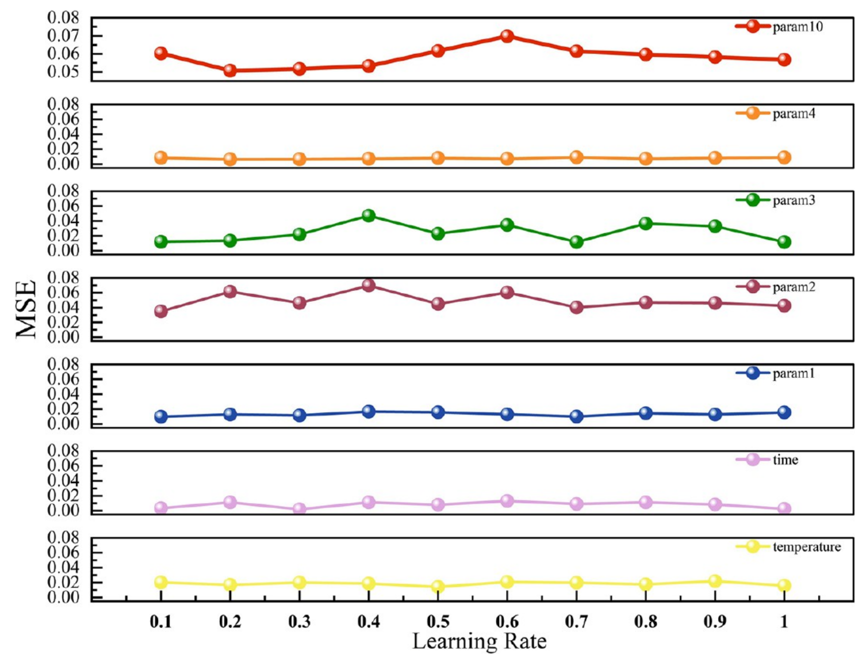


Figure 12. Assess the efficacy of the model by varying the learning rates and juxtaposing the mean squared error (MSE) values of distinct output parameters.

The learning rate is a hyperparameter that controls the extent to which we adjust the weights of our network and the loss gradient. The lower the value, the slower the slope decreases. Although this may be a good approach (using a low learning rate) to ensure that we do not miss any local minima, it may also mean that it takes us a long time to converge, especially if we become stuck in a stagnant area, as shown in Figure 12, which reflects the impact of the learning rate on various output features by comparing the MSE values in this study.

Figure 12 illustrates the effects of adjusting the model's learning rate, which ranges from 0.1 to 1, on various parameters. The optimal rates differ among the output parameters; for example, 0.5 is optimal for temperature, whereas for time, the optimal rates are 0.1 and 0.3. Varying rates are crucial for controlling predictions for individual outputs. Additionally, certain features such as param 2, param 3, and param 10 are more sensitive to rate changes, whereas parameters 1 and 4, along with temperature and time, remain relatively stable. Moreover, an appropriate learning rate can accelerate the optimization process of neural networks, aptly elucidating the optimal state between RAC_features and the synthesis conditions of MOFs. This, in turn, enhances the utilization of neural network models in the design and synthesis of MOFs, facilitating the discovery of optimal synthetic conditions.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

Table 4. Performance of Models with Different Hidden Nodes and Initial Populations[a]

IP	EI	NHLN 5	6	7	8	9	10	11	12	13	14	15	16	17
10	*R*	0.882	0.905	0.962	0.899	0.857	0.903	0.856	0.863	0.895	0.856	0.861	0.872	0.879
	MSE	0.047	0.026	0.034	0.034	0.031	0.034	0.039	0.034	0.030	0.033	0.033	0.030	0.036
20	*R*	0.912	0.905	0.913	0.898	0.840	0.868	0.839	0.891	0.901	0.869	0.580	0.851	0.868
	MSE	0.039	0.028	0.026	0.033	0.036	0.032	0.033	0.026	0.034	0.032	0.036	0.037	0.034
30	*R*	0.907	0.915	0.892	0.916	0.861	0.878	0.901	0.890	0.866	0.866	0.875	0.859	0.856
	MSE	0.049	0.027	0.265	0.029	0.031	0.032	0.033	0.032	0.041	0.035	0.035	0.034	0.031
40	*R*	0.867	0.885	0.903	0.849	0.869	0.846	0.852	0.880	0.870	0.883	0.853	0.893	0.846
	MSE	0.029	0.030	0.024	0.036	0.038	0.035	0.036	0.040	0.036	0.032	0.040	0.034	0.038
50	*R*	0.899	0.877	0.895	0.880	0.872	0.847	0.896	0.888	0.870	0.867	0.881	0.882	0.863
	MSE	0.026	0.028	0.028	0.030	0.038	0.033	0.033	0.038	0.036	0.033	0.034	0.033	0.037

[a]Note: NHLN is the number of hidden layer nodes. IP is the initial population. EI is the evaluation indicator.

We used genetic algorithms for optimization across various data sets. By defining decoding rules and fitness functions, we optimized neural network parameters, such as weights, biases, and structure. Adjusting the initial population size and hidden layer count helps fine-tune the model for optimal performance, as detailed in Table 4.

Table 4 and Figure 13 show that optimal performance is achieved when the initial population ranges from 10 to 40, and the number of hidden nodes is 6 or 7. Conversely, setting the initial population to 50 and having 5 hidden nodes result in better model performance.

Through genetic algorithm optimization, significant performance improvements were observed in the artificial neural networks. The *R* coefficient reached its peak at 0.962 with an initial population of 10 and 7 hidden nodes, surpassing the single neural network's *R* value of 0.86. Additionally, the error between the predicted and true values was small (RMSE = 3.1557), indicating close proximity between the predicted and true values. These enhancements demonstrate the ability of the optimized network to adapt to specific tasks, enhancing prediction accuracy and convergence speed.

In addition, by observing the parameter changes in the optimization process of the genetic algorithm, the key parameters, such as the network structure and connection weight, gradually tend to reach reasonable values during the optimization process, which indicates that the genetic algorithm can effectively search for the optimal parameter combination of the network.

4.3.2. Results of Model Validation. In the previous section, the existing data set was proportionally divided into a training set, a testing set, and a validation set, and the conclusion of internal validation was drawn that the model performance was good. Therefore, unused data from model building are used to evaluate the performance of the model on new data.

Training is performed according to the steps of the model establishment for the new data obtained. The original data are preprocessed, and the MRMR algorithm combined with an artificial neural network is used to select 123 feature inputs and 7 outputs. Finally, a genetic algorithm optimized artificial neural network is used to train and verify the selected feature inputs and outputs and to judge the performance of the model on new data, and the results are shown in Figure 14.

The performance of the new data on the model shows that it has good adaptability for new data, which indicates that predicting the synthesis conditions of the MOF on the basis of RAC_features is feasible and that the model has good generalizability.

4.3.3. Model Evaluation under Different Machine Learning Models. In this study, various machine learning models were compared to assess the efficacy of the proposed model in predicting the MOF synthesis conditions. The Pearson coefficient and MSE were utilized for model effectiveness comparisons.

In this study, trees (fine trees, medium trees, and optimized trees), linear regression, SVM (linear SVM, fine Gaussian SVM), GPR (quadratic rational GPR and square exponential GPR), artificial neural networks (ANNs), and artificial neural networks optimized by a genetic algorithm (GANN) were used to compare the *R* values of these models. The results of the comparative analysis of the selected models after training are shown in Figure 15.

A comparative analysis of Figure 15 highlights the performance of selected regression models: ANN ($R = 0.86$), GANN ($R = 0.96$), quadratic rational GPR ($R = 0.84$), and squared exponential GPR ($R = 0.776$). Notably, ANN and GANN exhibit strong performance, with GANN showing a significant advantage. Consequently, the GANN algorithm was chosen for predicting the MOF synthesis conditions in this study.

4.4. Impact of Different Input Parameters on the Output Varies Across Categories. In the original data set, we observed distinct classifications. After training the model to determine the optimal input and output parameters, we evaluated the model performance by categorizing the data into input parameter groups. From the observed data, 123 input points were classified into seven categories. To assess the impact of each category on model performance, we systematically removed one category at a time and compared the *R* values and MSEs to observe the effect. Figure 16 illustrates the model performance after training.

Figure 16 shows that eliminating specific input features impacts the model's performance. Notably, when the chi class input is removed, the minimum *R* value is 0.81078, and the MSE value is relatively large at 0.15014. This significant impact is visually represented by the corresponding bar graph reaching its lowest point. In contrast, excluding inputs from classes T, I, Z, and geo results in peak values on the bar graph, indicating their minimal influence on the model. This finding implies that the variation in chi-class RAC_features has the greatest impact on the prediction of synthesis conditions for MOFs; immediately following are the S-class and alpha-class data. The removal of these compounds has a significant effect on the model. Researchers should pay more attention to the input features that have a significant influence on the synthesis conditions when synthesizing MOFs.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

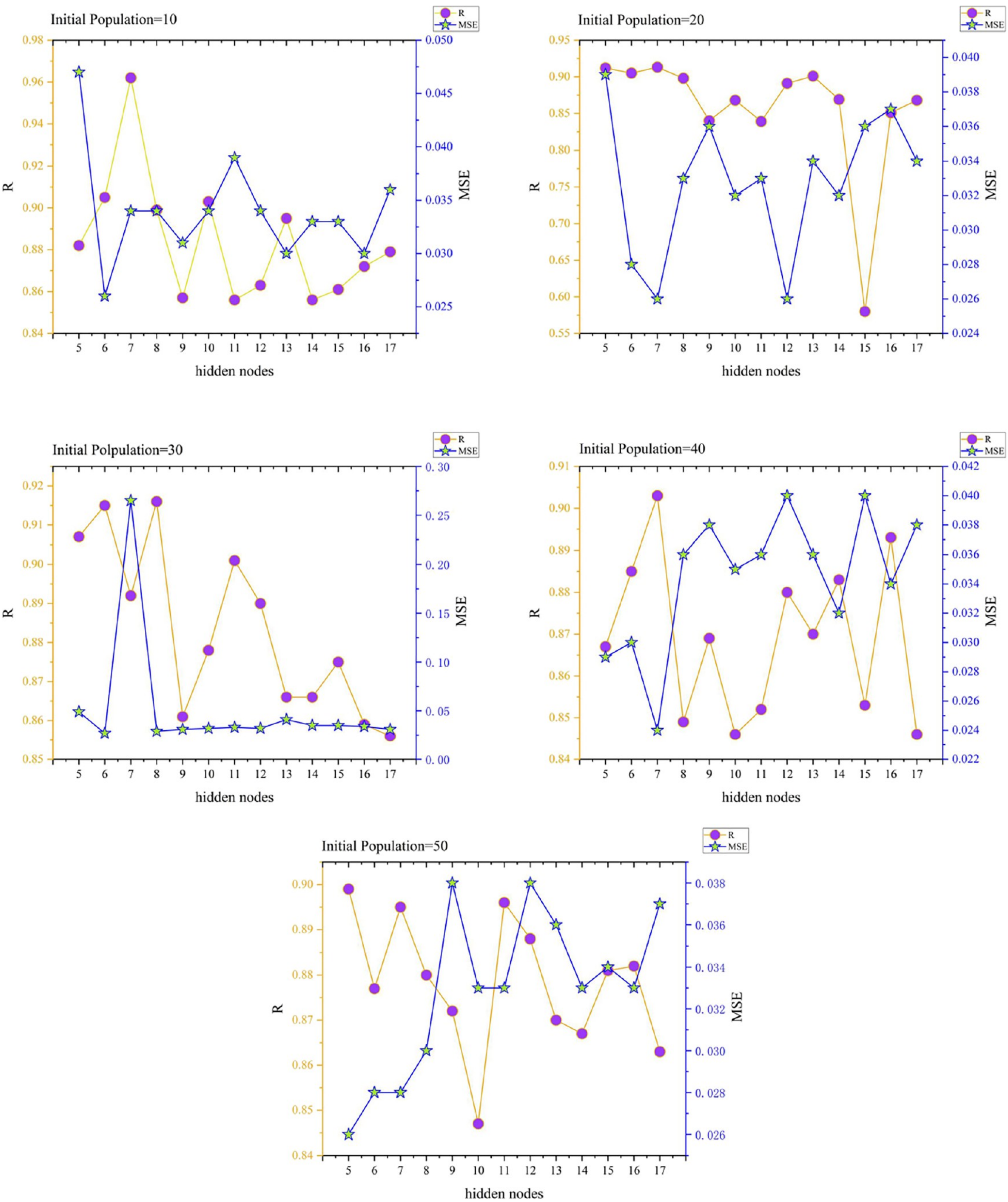


Figure 13. Impact of varying the number of hidden layer nodes and initial population sizes on the model.

5. DISCUSSION AND CONCLUSIONS

This study utilized a BP neural network optimized by a genetic algorithm to predict MOF synthesis conditions and investigate related parameters. The aim was to determine the optimal synthesis conditions for reducing material and financial waste for manufacturing enterprises. The contributions of this research include the following:

1. When analyzing individual output parameters, we find minimal correlation between certain data and input features in the raw data. Consequently, these less

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100

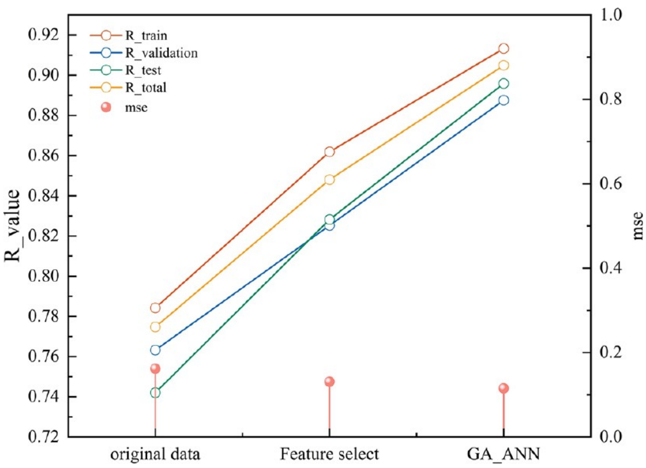


Figure 14. Display of validation results of new data on the established model.

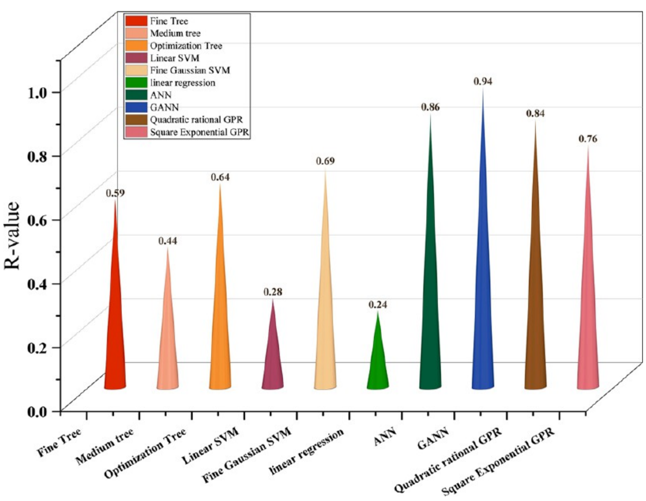


Figure 15. Judging the performance of different models by comparing the Pearson correlation coefficient (R) values.

influenced output parameters can be omitted in further analysis. The 123 selected input features were selected to enhance the model performance.

2. When adjusting the model parameters, analysis of the input feature importance scores revealed categories with higher averages, indicating their greater influence on predicting synthesis conditions. These factors should be prioritized in production preparation. The varying impacts of learning rates suggest that manufacturers can use optimal parameters for timely production adjustments. The best model performance is achieved with an initial population size of 10 and 7 nodes in the hidden layer.
3. In metal material research, MOFs have received limited attention, especially in terms of their ability to predict synthesis conditions. Therefore, when analyzing the same data set, the GANN model proposed in this paper exhibits significantly superior performance in comparison to traditional neural networks, support vector machines (SVMs), Gaussian process regression (GPR), and tree-based methods. This enhanced performance can be attributed to the unique architecture and training methodology of the GANN model, which allows it to better capture complex patterns and relationships within the data.
4. MOFs offer vast application potential because of their controllable pore structures and expansive surface areas and are ideal for applications such as H_2 adsorption, catalysis, and the storage of fuels such as methane and hydrogen. The chi input features and alpha input features are electronegativity and polarizability-related features that affect the ability of adsorbents to retain molecules. These characteristics also affect the reactivity and stability, which are crucial for the synthetic prediction.

Many material manufacturing enterprises struggle to achieve large-scale production due to material characteristics and costs, resulting in a significant waste of funds and resources. This paper proposes a method for precise prediction in MOF production research, aiming to enable intelligent manufacturing and reduce waste. Similar challenges exist in the production of biomaterials such as COFs, porous polymers, bioceramics, and bioglass, suggesting the potential extension of this method to their synthesis and application, thus reducing waste and enabling intelligent manufacturing.

This prediction model shows high accuracy in forecasting MOF synthesis conditions using existing data but faces

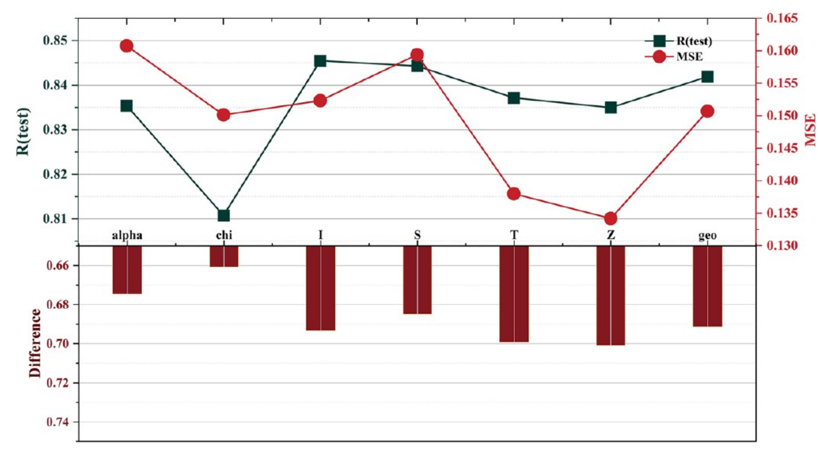


Figure 16. Performance of a model if analyzed by removing data of the same category. The evaluation criteria observed are the model's R value and MSE value.

limitations such as potential biases when predicting new conditions due to reliance solely on these data and the unpredictable nature of MOF experiments.

In conclusion, this study offers a viable method for predicting MOF synthesis conditions, which was validated through accurate experiments. This methodology also shows promise for the intelligent manufacturing of other materials, providing valuable insights for future research and the design of MOFs.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are openly available at https://github.com/aimat-lab/MOF_Synthesis_Prediction/tree/main/code/RAC_features

■ AUTHOR INFORMATION

Corresponding Author

Guangying Jin − *School of Maritime Economics and Management, Dalian Maritime University, Dalian 116026, China*; orcid.org/0000-0002-4459-8425; Email: guangying.jin@dlmu.edu.cn

Authors

Wei Ran − *School of Maritime Economics and Management, Dalian Maritime University, Dalian 116026, China*

Manyue Zhang − *School of Maritime Economics and Management, Dalian Maritime University, Dalian 116026, China*

Yun Li − *School of Maritime Economics and Management, Dalian Maritime University, Dalian 116026, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jcim.4c00997

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Funding

This work was supported by the Fundamental Research Funds for the Central Universities, grant number 3132022291.

Notes

The authors declare no competing financial interest.

■ REFERENCES

(1) van Eeckhout, J.; Jacobs, C. K. Synthesis, structure, and properties of molecular sieves built with metal atoms polyhedra used as building blocks. *J. Solid State Chem.* **1989**, *78* (1), 215−220.

(2) Ding, M.; Cai, X.; Jiang, H. L. Improving MOF stability: approaches and applications. *Chem. Sci.* **2019**, *10* (44), 10209−10230.

(3) Zhang, X.; Chen, A.; Zhong, M.; Zhang, Z.; Zhang, X.; Zhou, Z.; Bu, X. H. Metal−Organic Frameworks (MOFs) and MOF-Derived Materials for Energy Storage and Conversion. *Electrochem. Energy Rev.* **2018**, 29.

(4) Du, P.; Dong, Y.; Liu, C.; Wei, W.; Liu, D.; Liu, P. Fabrication of hierarchical porous nickel based metal-organic framework (Ni-MOF) constructed with nanosheets as novel pseudo-capacitive material for asymmetric supercapacitor. *J. Colloid Interface Sci.* **2018**, *518*, 57−68.

(5) Manikandan, M. R.; Cai, K. P.; Hu, Y. D.; Li, C. L.; Zhang, J. T.; Zheng, Y. P.; Liang, Y. F.; Song, H. R.; Shang, M. Y.; Shi, X. N.; Zhang, J. X.; Yin, S. Q.; Shang, S. Y.; Wang, X. W. Influence of hydrothermal reaction time on the supercapacitor performance of NiMOF nanostructures. *Appl. Phys. A* **2021**, *127*, 421.

(6) Li, Y.; Xu, Y.; Liu, Y.; Pang, H. Exposing {001} crystal plane on hexagonal Ni-MOF with surface-grown cross-linked mesh-structures for electrochemical energy storage. *Small* **2019**, *15*, No. 1902463.

(7) Yang, C.; Li, X.; Yu, L.; Liu, X.; Yang, J.; Wei, M. A new promising Ni-MOF superstructure for high-performance supercapacitors. *Chem. Commun.* **2020**, *56*, 1803−1806.

(8) Zhang, C.; Zhang, Q.; Zhang, K.; Xiao, Z.; Yang, Y.; Wang, L. Facile synthesis of a two dimensional layered Ni-MOF electrode material for high performance supercapacitors. *RSC Adv.* **2018**, *8*, 17747−17753.

(9) Yang, J.; Xiong, P.; Zheng, C.; Qiu, H.; Wei, M. Metal−organic frameworks: a new promising class of materials for a high performance supercapacitor electrode. *J. Mater. Chem. A* **2014**, *2* (39), 16640−16644.

(10) Lee, Y. H.; Kwon, Y.; Kim, C.; Hwang, Y. E.; Choi, M.; Park, Y.; Jamal, A.; Koh, D. Y. Controlled Synthesis of Metal−Organic Frameworks in Scalable Open-Porous Contactor for Maximizing Carbon Capture Efficiency. *JACS Au.* **2021**, 1198.

(11) McHugh, L. N.; McPherson, M. J.; McCormick, L. J.; Morris, S. A.; Wheatley, P. S.; Teat, S. J.; McKay, D.; Dawson, D. M.; Sansome, C. E. F.; Ashbrook, S. E.; Stone, C. A.; Smith, M. W.; Morris, R. E. Hydrolytic stability in hemilabile metal−organic frameworks. *Nat. Chem.* **2018**, *10*, 1096−1102.

(12) Ding, R.; Wang, R.; Ding, Y.; Yin, W.; Liu, Y.; Li, J.; Liu, J. Designing AI-Aided Analysis and Prediction Models for Nonprecious Metal Electrocatalyst-Based Proton-Exchange Membrane Fuel Cells. *Angew. Chem.* **2020**, *132* (43), 19337−19345.

(13) Haldar, D.; Shabbirahmed, A. M.; Mahanty, B. Multivariate regression and artificial neural network modelling of sugar yields from acid pretreatment and enzymatic hydrolysis of lignocellulosic biomass. *Bioresour. Technol.* **2023**, *370*, No. 128519.

(14) Kurani, A.; Doshi, P.; Vakharia, A.; Shah, M. A Comprehensive Comparative Study of Artificial Neural Network (ANN) and Support Vector Machines (SVM) on Stock Forecasting. *Ann. Data Sci.* **2023**, *10*, 183−208.

(15) Darwan, D.; Lim, K. R. G.; Wijaya, H.; Lim, Z. C.; Wang, T.; Ang, W. H.; Tan, Z. Deep Fluorescence Imaging by Laser-Scanning Excitation and Artificial Neural Network Processing. *Adv. Opt. Mater.* **2020**, *8* (19), No. 2000390.

(16) Vo, N. D.; Oh, D. H.; Kang, J. H.; Oh, M.; Lee, C. H. Dynamic-model-based artificial neural network for H2 recovery and CO2 capture from hydrogen tail gas. *Appl. Energy* **2020**, *273*, No. 115263.

(17) Zhan, Y.; Zhu, J.; Schrader, L. C.; Wang, D. Modeling and optimization of bioenergy production from co-digestion of poultry litter with wheat straw in anaerobic sequencing batch reactor: Response surface methodology and artificial neural network. *Appl. Energy* **2023**, *345*, No. 121373.

(18) Gu, C.; Wang, X.; Song, Q.; Li, H.; Qiao, Y. Prediction of gas-liquid-solid product distribution after solid waste pyrolysis process based on artificial neural network model. *Int. J. Energy Res.* **2021**, *45* (9), 13786−13800.

(19) Timoshenko, J.; Haase, F. T.; Saddeler, S.; Rüscher, M.; Jeon, H. S.; Herzog, A.; Hejral, U.; Bergmann, A.; Schulz, S.; Roldan Cuenya, B. Deciphering the Structural and Chemical Transformations of Oxide Catalysts during Oxygen Evolution Reaction Using Quick X-ray Absorption Spectroscopy and Machine Learning. *J. Am. Chem. Soc.* **2023**, *145* (7), 4065−4080.

(20) Kotobi, A.; Singh, K.; Höche, D.; Bari, S.; Meißner, R. H.; Bande, A. Integrating Explainability into Graph Neural Network Models for the Prediction of X-ray Absorption Spectra. *J. Am. Chem. Soc.* **2023**, *145* (41), 22584−22598.

(21) Illarionov, A.; Sakipov, S.; Pereyaslavets, L.; Kurnikov, I. V.; Kamath, G.; Butin, O.; Voronina, E.; Ivahnenko, I.; Leontyev, I.; Nawrocki, G.; Darkhovskiy, M.; Olevanov, M.; Cherniavskyi, Y. K.; Lock, C.; Greenslade, S.; Sankaranarayanan, S. K.; Kurnikova, M. G.; Potoff, J.; Kornberg, R. D.; Levitt, M.; Fain, B. Combining Force Fields and Neural Networks for an Accurate Representation of Chemically Diverse Molecular Interactions. *J. Am. Chem. Soc.* **2023**, *145* (43), 23620−23629.

(22) Lin, J.; Liu, Z.; Guo, Y.; Wang, S.; Tao, Z.; Xue, X.; Li, R.; Feng, S.; Wang, L.; Liu, J.; Gao, H.; Wang, G.; Su, Y. Machine learning accelerates the investigation of targeted MOFs: Performance prediction, rational design and intelligent synthesis. *Nano Today* **2023**, *49*, No. 101802.
(23) Raccuglia, P.; Elbert, K. C.; Adler, P. D. F.; Falk, C.; Wenny, M. B.; Mollo, A.; Zeller, M.; Friedler, S. A.; Schrier, J.; Norquist, A. J. Machine-learning-assisted materials discovery using failed experiments. *Nature* **2016**, *533*, 73−76.
(24) Xie, Y.; Zhang, C.; Hu, X.; Zhang, C.; Kelley, S. P.; Atwood, J. L.; Lin, J. Machine Learning assisted synthesis of metal-organic nanocapsules. *J. Am. Chem. Soc.* **2020**, *142*, 1475−1481.
(25) Zhang, Q.; Liang, H.; Tao, Y.; Yang, J.; Tang, B.; Li, R.; Ma, Y.; Ji, l.; Jiang, X.; Li, S. Size-Controllable Eu-MOFs through Machine Learning Technology: Application for High Sensitive Ions and Small-Molecular Identification. *Small Methods* **2022**, *6*, No. 2200208.
(26) Luo, Y.; Bag, S.; Zaremba, O.; Cierpka, A.; Andreo, J.; Wuttke, S.; Friederich, P.; Tsotsalas, M. MOF Synthesis Prediction Enabled by Automatic Data Mining and Machine Learning. *J. Ger. Chem. Soc.* **2022**, No. e202200242.
(27) Le, T. C.; Winkler, D. A. Discovery and optimization of materials using evolutionary approaches. *Chem. Rev.* **2016**, *116*, 6107−6132.
(28) Moosavi, S. M.; Chidambaram, A.; Talirz, L.; Haranczyk, M.; Stylianou, K. C.; Smit, B. Capturing chemical intuition in synthesis of metal-organic frameworks. *Nat. Commun.* **2019**, *10*, 539.
(29) Zhang, Liet al.Challenges and strategies for sustainable synthesis of metal-organic frameworks. *Nat. Rev. Mater.* (**2020**).
(30) Yang, Liu,et al.Reducing waste in the synthesis of metal-organic frameworks: A critical review. *Chem. Soc. Rev.* (**2019**).
(31) *Karlsruhe Institute of Technology* (2022). Retrieved from [https://www.kit.edu/].
(32) Nandy, A.; Terrones, G.; Arunachalam, N.; Duan, C.; Kastner, D. W.; Kulik, H. J. MOFSimplify, machine learning models with extracted stability data of three thousand metal−organic frameworks. *Sci. Data* **2022**, *9*, 74.
(33) McCulloch, W. S.; Pitts, W. A Logical Calculus of the Ideas Immanent in Nervous Activity. *Bull. Math. Biophys.* **1943**, *5*, 115−133.
(34) Mütter, F.; Berger, C.; Königshofer, B.; Höber, M.; Hochenauer, C.; Subotić, V. Artificial intelligence for solid oxide fuel cells: Combining automated high accuracy artificial neural network model generation and genetic algorithm for time-efficient performance prediction and optimization. *Energy Convers. Manage.* **2023**, *291*, No. 117263.
(35) Singh, V.; Mehra, R.; Ramesh, K. B.; Srivastava, P.; Mishra, A. Treatment of carpet and textile industry effluents using Diplosphaera mucosa VSPA: A multiple input optimization study using artificial neural network-genetic algorithms. *Bioresour. Technol.* **2023**, *387*, No. 129619.

https://doi.org/10.1021/acs.jcim.4c00997
J. Chem. Inf. Model. 2025, 65, 1085−1100